

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) is an increasingly general route for improving the reasoning behavior of large language models whenever task outcomes can be checked automatically. Yet widely used policy-optimization variants such as GRPO can lose useful gradients under conditions that arise broadly in verifiable-reward training: token-level probability ratios are clipped by fixed bounds, and responses with identical rewards are standardized to zero advantage. These design choices can weaken updates and leave generated responses underused even when they contain informative reasoning variation. We propose **Dynamic Clipping Policy Optimization (DCPO)**, a policy-optimization framework that keeps the same RLVR setting while making the update rule more responsive to the probability structure of generated tokens and the reward history of each prompt. DCPO combines a dynamic clipping rule that adapts bounds from token-specific prior probabilities with a smooth advantage standardization rule that incorporates cumulative training-step statistics, thereby broadening exploration at the token level and improving response-level use of generated data. Across four mathematical reasoning benchmarks and four model settings, DCPO achieves state-of-the-art results under the evaluated conditions. On Qwen2.5-Math-7B, DCPO reaches 46.7 Avg@1 with greedy decoding and 38.8 Avg@32 with 32-sample evaluation on AIME24, exceeding DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On Qwen2.5-14B for AIME25, DCPO obtains (23.3/19.0), compared with GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). DCPO also improves the nonzero-advantage ratio by an average of 28% over GRPO across four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude compared with GRPO and DAPO while maintaining stronger performance. These findings indicate that probability-aware clipping and cumulative advantage smoothing can make RLVR updates more data-efficient across the evaluated model scales and benchmark conditions.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central focus across tasks that require multi-step inference, search, and self-correction (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) offers a broadly useful training paradigm for such settings because it optimizes models with rule-based outcome feedback rather than relying on subjective preference labels. This makes RLVR especially natural for domains where answers can be checked automatically, including mathematics and coding, while still reflecting a general pattern of verifiable decision-making in LLM training (DeepSeek-AI et al., 2024).

The GRPO family of algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) has become a common foundation for applying RLVR to LLMs. Its empirical strength has been reported across several studies, but recent analyses such as DAPO (Yu et al., 2025) show that the same design can encounter limitations that are not tied to a single benchmark. In particular, GRPO can exhibit entropy collapse, with advantage values decaying to zero in later training, and its fixed clipping behavior can suppress learning signals from rarely sampled tokens. These effects narrow policy exploration and can reduce the benefit of generated responses in any RLVR setting where correct outcomes are sparse or unevenly distributed.

DAPO addresses part of this broader failure mode through Clip-Higher, which relaxes the upper clipping bound, and Dynamic Sampling, which removes samples with uniform rewards so that each

batch contains valid gradients. These changes improve over GRPO in important cases, but they leave two general sources of inefficiency. First, **Response-Level Inefficiency**: filtering responses lowers sampling efficiency and can slow convergence because generated data are discarded before optimization. Second, **token clipping deficit**: our analysis shows that useful clipping ranges depend on token-specific probabilities, so a single relaxed upper bound remains suboptimal across the token distribution. Even approaches that avoid directly discarding zero-gradient responses still face these broader response-utilization and token-level exploration issues.

We introduce **Dynamic Clipping Policy Optimization (DCPO)** to address these limitations without changing the underlying RLVR task, data, or reward setting. First, DCPO uses a **smoothing technique that standardizes advantages across cumulative steps**, preventing responses with identical current-step rewards from automatically losing all gradient signal. By combining the reward distribution accumulated for a prompt with the distribution observed at the current step, this mechanism supports more efficient response-level exploration across repeated generations (see Section 2). Second, DCPO introduces a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities**. Rather than choosing different fixed bounds for all tokens, DCPO widens the allowable ratio range for lower-probability tokens and keeps tighter updates for common tokens, providing probability-aware token-level exploration (see Section 2 and Appendix A.6). In addition, DCPO computes the mean loss within each response instead of averaging over the full batch, preserving the relative advantage structure among responses to the same prompt and avoiding dilution from batch-level aggregation.

For empirical evaluation, we keep the study within a controlled mathematical-reasoning RLVR setting: the training data merge Math-DAPO-17k (Yu et al., 2025) with levels 3-5 of the MATH dataset (Liu et al., 2025), and evaluation uses four standard mathematical reasoning benchmarks. We test four model settings and compare with GRPO, DAPO, and, where available, GSPO. Within this scope, DCPO consistently performs best, with the largest gains on the more challenging AIME benchmarks: on AIME24-Avg@32 with 32-sample evaluation, DCPO-7B scores 38.8 compared with 32.1 (GRPO), 31.6 (DAPO), and 34.9 (GSPO), while on AIME25-Avg@32, DCPO-14B reaches 19.0 compared with 10.5 (GRPO), 15.3 (DAPO), and 9.9 (GSPO). DCPO also increases the average nonzero advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with the baselines while delivering stronger benchmark performance.

Additional background on the loss and advantage calculations used by GRPO, DAPO, and GSPO is provided in Appendix A.2, so the main text focuses on the parts of the optimization pipeline that generalize across the evaluated RLVR updates.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Probability-ratio clipping is a general stabilization device in reinforcement learning: it limits policy updates so that optimization does not move too far from the behavior distribution that produced the data. This consideration is especially relevant for importance-weighted objectives, including the token-level updates used in RLVR. When importance sampling is reweighted by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$, the estimate of $\mathbb{E}_{q(x)}[f(x)\frac{p(x)}{q(x)}]$ remains unbiased, but its variance can grow substantially, as shown in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right)^2 \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right)^2 p(x) dx \quad (1)$$

To control this variance in a simple and broadly deployable way, prior work (e.g., Schulman et al. (2017)) constrains the probability ratio with a fixed clipping hyperparameter ϵ , as in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed bound, however, treats all tokens as if they had the same prior probability, which narrows exploration for low-probability regions of the policy. In such regions, the same relative bound translates into a limited absolute update range, reducing the ability to learn from rare but informative tokens. Asymmetric variants such as DAPO relax one side of the bound, but the constraint is still global rather than probability-dependent. GSPO takes a sequence-level view and discards entire

responses when the sequence-level importance ratio exceeds asymmetric bounds, which can remove informative tokens while retaining other high-variance tokens and thereby affect stability.

From the perspective of balancing variance control with exploration, a more general constraint should depend on the probability of the token being updated, as formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

By substituting $p(x) = r(x)q(x)$ into Equation (3) and enforcing non-negativity of the square-root term through a maximum with zero, we obtain a closed-form range for $r(x)$. The resulting **dynamic-adaptive clipping bounds** vary with the old probability $q(x)$ and with hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This dynamic-adaptive clipping rule gives low-probability tokens more room to change while retaining conservative bounds for high-probability tokens, making the update less tied to a single global exploration scale. Following the motivation of dual clipping (Ye et al., 2020), we cap the maximum bound for both positive and negative advantages at 10 to avoid overly large importance weights and overtraining. Appendix A.5 and Appendix A.6 provide the derivation and additional analysis from the importance-sampling perspective.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

In GRPO, DAPO, GSPO, and related RLVR methods, the advantage $\hat{A}_{new,j}^i$ for the j -th response is standardized using only the rewards $R_{j=1,\dots,G}^i$ generated for the same prompt at step i , as shown in Equation (11) in Appendix A.3. This local standardization is simple, but it creates issues that can appear in many verifiable-reward tasks. If sampling randomness yields identical rewards at one step, every advantage becomes zero, so the prompt contributes no update even when its reasoning trajectories differ. In addition, high-entropy sampling can produce highly skewed label counts, causing standardized advantages to fluctuate sharply across steps and even change sign, which destabilizes training.

Recent work (Yue et al., 2025) further suggests that RLVR often refines an existing policy by changing the likelihood of correct samples rather than replacing the full response distribution. Under this view, responses to the same prompt across training steps can be treated as samples from a broader prompt-level reward distribution, motivating the *cumulative* standardization in Equation (5).

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the statistics aggregate all responses generated so far for the same prompt.

To reduce the instability of both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we define two smoothed quantities, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$. They are weighted averages of the current-step and cumulative advantages, with weights that evolve over step i as in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

$\hat{S}A_{new,j}^i$ increasingly emphasizes the current distribution as training advances, allowing the update to remain responsive to the latest policy outputs. Conversely, $\hat{S}A_{total,j}^i$ increasingly emphasizes the cumulative distribution, interpreting the current-step deviation within a broader prompt-level history.

The final advantage $\hat{A}_j^i$ is the smoothed value with smaller absolute magnitude, which limits the effect of fluctuations from either current-step or cumulative standardization and is defined in Equation (7).

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

After a prompt has entered optimization, later responses for that prompt can continue to influence updates even when the current-step advantage would otherwise be zero. Thus, when identical rewards occur within a step, those responses still update the model through the cumulative component $\frac{1}{i}\hat{A}_{total,j}^i$, preserving learning signal from generated data and improving response-level efficiency.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** (the loss Equation (13) used in DAPO) equalizes token contributions, but this token-centric aggregation can distort the intended relationship among responses of different lengths. Consider two responses to the same prompt: response A has advantage 1 and length 500, while response B has advantage 0.5 and length 1500. Under TLM, A receives weight $\frac{500}{500+1500} = 0.25$ and B receives weight $\frac{1500}{500+1500} = 0.75$. Their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, so the lower-advantage response has greater influence solely because it is longer.

Batch-level averaging creates a complementary issue. The **Sequence-Level Mean loss (SLM)** (the loss Equation (12) used in GRPO) weakens advantage magnitudes after advantage calculation, effectively reducing them by a factor of G . With large batches, this can blur the standardized relationships that were computed among responses to the same prompt. Because standardized advantages already encode relative response quality, we average over tokens within each response using $|o_i|$ and do not average across all responses in the batch. This **Only Token Mean loss (OTM)**, defined in Equation (8), preserves response-level comparisons while assigning equal token-level weight within a response.

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, the DCPO loss is not constrained by a KL divergence term. Its lower and upper clipping limits are the dynamic bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate DCPO across four Qwen model settings that differ in scale and pretraining specialization: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). The training set is a merged collection of approximately 25k mathematical problems, combining DAPO-Math-17K (Yu et al., 2025) with levels 3-5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). All problems use the Qwen-Math template for both training and evaluation (see Section A.8), keeping the prompt format fixed across model classes and benchmarks.

Evaluation We evaluate on four widely used mathematical reasoning benchmarks that span standard problem solving and competition-style difficulty: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). For **MATH500**, we report only Avg@1 because it contains 500 problems. For the smaller **AMC23** (40 problems), **AIME24** (30 problems), and **AIME25** (30 problems) benchmarks, we report both Avg@1 and Avg@32 so the evaluation captures deterministic accuracy and sampled-policy behavior.

- **Avg@1:** This metric is the standard greedy-decoding accuracy and reflects the model’s single best deterministic prediction.
- **Avg@32:** This metric averages accuracy over 32 sampled responses per problem using temperature 1.0 and top.p 1.0, characterizing the robustness and stability of the trained policy distribution under sampling.

Baseline Settings We use GRPO and DAPO as primary baselines and include GSPO for Qwen2.5-Math-7B and Qwen2.5-14B where those comparisons are available. Other implementation details, including clipping thresholds ϵ_* , the codebase, and shared training hyperparameters, are reported in Appendix A.9.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1: Performance on the evaluated mathematical reasoning benchmarks. Avg@1 reports greedy decoding and Avg@32 reports 32-sample decoding; bold values mark the best result in each comparison, and Δ gives the change relative to each baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/+0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/+0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/+2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/+2.2	+6.6/+10.1	+4.2/+5.5

Table 1 compares GRPO, DAPO, GSPO, and DCPO on MATH500, AMC23, AIME24, and AIME25. Reporting both greedy decoding (Avg@1) and sampled decoding (Avg@32) lets us evaluate whether each optimization method improves not only a single deterministic output but also the broader response distribution. Across the four evaluated model settings, DCPO is consistently superior or competitive with GRPO, DAPO, and, where available, GSPO, supporting its usefulness as a probability-aware RLVR optimization method within this mathematical reasoning scope.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO obtains the highest average scores (47.2/32.8), improving over GRPO (+1.2/+0.8) and DAPO (+0.7/+0.4). On Qwen2.5-3B, DCPO reaches (37.6/22.7), exceeding GRPO (+1.3/+1.7) and remaining competitive with DAPO,

with a higher Avg@1 (+1.0) and a small Avg@32 gap (-0.4). For Qwen2.5-Math-7B, DCPO achieves (57.1/45.2), surpassing GRPO (+4.0/+3.6), DAPO (+3.3/+2.8), and GSPO (+1.9/+1.9). On Qwen2.5-14B, the largest tested model, DCPO reaches (53.2/39.0), improving over GRPO (+6.5/+7.7), DAPO (+1.3/+0.1), and GSPO (+4.2/+5.5). These results show that the same DCPO update remains effective across the tested capacities and supports both greedy and stochastic decoding performance.

Robustness on AIME under Sampling (Avg@32) The AIME24 and AIME25 results are especially informative because these competition-style benchmarks stress sampled reasoning under harder conditions. Under Avg@32, which measures whether the trained policy can produce correct answers across high-entropy samples, DCPO is strongest at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B scores 38.8, compared with GRPO (32.1), DAPO (31.6), and GSPO (34.9); on AIME25-Avg@32, DCPO-14B reaches 19.0, compared with GRPO (10.5), DAPO (15.3), and GSPO (9.9). These gains indicate that DCPO improves the reliability of correct generations across sampled outputs, a property that matters whenever downstream use relies on multiple candidate responses rather than a single greedy answer.

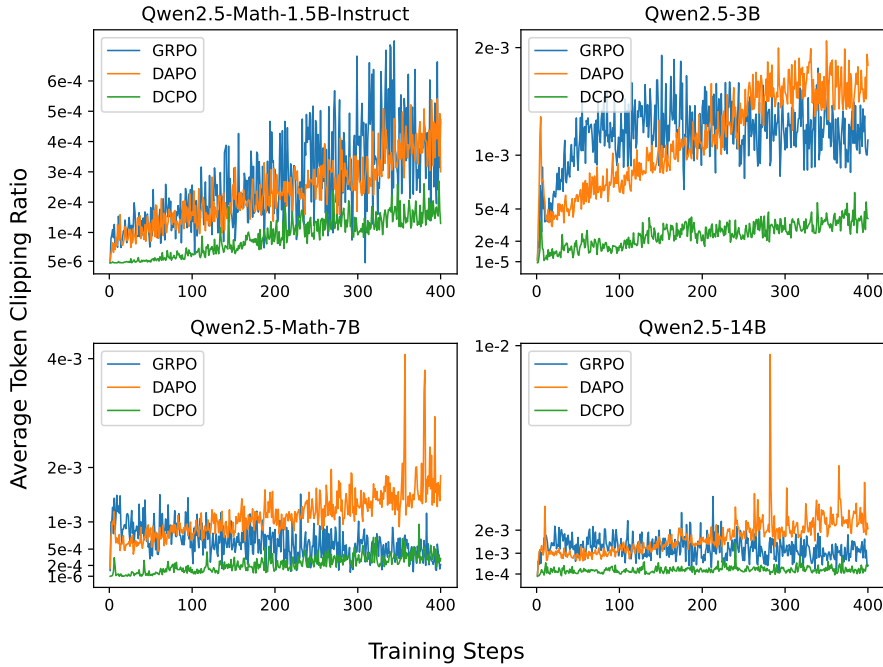


Figure 1: Token clipping ratio across the evaluated models and optimization methods.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio (TCR)** as the proportion of generated tokens excluded from policy updates during back-propagation because their ratios are clipped. As Figure 1 shows, this diagnostic differs substantially across optimization methods and therefore reveals how much token-level signal each method retains.

$$\text{TCR} = \text{Average} \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } \textit{micro}_m}{\text{Total number of tokens in } \textit{micro}_m} \right) \quad (9)$$

where $\textit{micro}_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

For GRPO, TCR trends differ by model scale: it rises over training for Qwen2.5-Math-1.5B and Qwen2.5-3B, but gradually declines for Qwen2.5-Math-7B and Qwen2.5-14B. DAPO instead shows an increasing TCR across all evaluated models, meaning that more tokens are excluded as training progresses and more policy updates rely on partial or truncated response information. Both GRPO and DAPO also show large fluctuations and occasional spikes, which may introduce instability and

contribute to entropy collapse. GSPO has still higher average *TCR* in the evaluated settings, exceeding 11% on Qwen2.5-Math-7B and 15% on Qwen2.5-14B, so its sequence-level clipping can remove more informative token signal than token-level clipping methods.

DCPO shows a more stable token-retention profile. Across model sizes and training stages, its *TCR* remains relatively constant and about an order of magnitude lower than GRPO and DAPO. This means that fewer generated tokens are discarded, so more of each response can contribute to the update while rare-token exploration is still bounded by the dynamic clipping rule. The result supports the broader claim that probability-aware clipping improves sample efficiency without allowing unbounded outlier variation.

Qwen2.5-Math-7B illustrates why the aggregate *TCR* can remain low even when clipping matters for exploration. The token distribution rapidly concentrates on high-confidence tokens: after roughly 60 steps, about 95% of generated tokens have probability ($q(x) > 0.9$), rising to about 97% after roughly 100 steps and continuing upward later in training. As shown in Figure 4d, once the old probability satisfies ($q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83$), the new probability $p(x)$ participates in updates within $[\frac{1}{1+\epsilon}, 1]$ without being discarded by the clipping methods. Most tokens therefore occupy a region that is not clipped, explaining the low overall *TCR*. At the same time, the later-stage increases in *TCR* in Figure 1 show that low-probability tokens are increasingly at risk of being discarded, precisely where exploration diversity is most important.

Prior work (Wang et al., 2025) shows that **high-entropy tokens**, which correspond to low-probability choices, are important drivers of emergent reasoning behavior. Since lower-probability tokens carry more uncertainty and often more information, giving them a broader admissible update range expands the effective exploration space available to the policy and can support more diverse reasoning trajectories.

These empirical and theoretical observations motivate **dynamic-adaptive clipping**: it relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the safe interval for high-confidence tokens. By expanding the update interval in the high-entropy tail, DCPO allows the model to learn from rare informative tokens without changing the treatment of common tokens, which helps improve reasoning performance under the evaluated RLVR conditions.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

During RLVR training, a generated response contributes to learning only when it produces a nonzero gradient. We quantify this response-level efficiency with the **Response Utilization Ratio (*RUR*)**, defined as the percentage of generated responses with nonzero advantage that participate in policy updates.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 shows that the average *RUR* for GRPO is low across the four models, with an overall mean of 44.6%. Thus, more than half of generated responses fail to provide update signal, limiting response-level exploration. DCPO, by contrast, reaches an average *RUR* of approximately 70% after the first epoch, corresponding to an absolute increase of 28% and a relative improvement of 64% over GRPO.

Table 2: Average response utilization ratio over 400 training steps.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* declines substantially over time across model sizes and model types, dropping from above 90% to below 50%. On Qwen2.5-Math-7B, for example, *RUR* falls to roughly 30% after about 200 training steps. This pattern indicates that current-step advantage standardization creates many zero-gradient responses, reducing exploration efficiency later in training. DCPO’s smooth advantage standardization maintains a much higher *RUR*: after the first epoch, it stabilizes near 70% and continues to improve with further training.

The *RUR* exceeds 90% during the earliest steps, then remains around 70% from the second epoch through the end of training rather than approaching 100%. This behavior clarifies what SAS contributes and why some responses should still remain unused.

First, the early *RUR* above 90% is largely caused by immature model outputs that fail to follow the required format, receive negative rewards, and inflate the utilization statistic. This suggests that models rapidly learn fixed answer patterns, after which training should emphasize reasoning improvement rather than format acquisition. Second, two factors explain why zero advantages persist and why *RUR* stabilizes around 70% rather than 100%: (i) nearly half of the math problems are simple enough that the model consistently solves them correctly, making further policy updates unnecessary for those instances; (ii) other instances remain consistently wrong, either because they are too difficult for the current model even with correct labels or because they contain label noise. The DAPO-Math-17K dataset also contains some items whose labels were converted to mislabeled integers, introducing potentially incorrect labels. In these cases, discarding zero-advantage responses acts as a filter that reduces the risk of updating the model toward noisy or mislabeled data.

DAPO discards generated responses with identical labels, unlike GRPO and DCPO, so a direct *RUR* comparison is not meaningful. We therefore evaluate DAPO’s training efficiency by measuring the number of generated responses and training time required to complete the same number of update steps. To compensate for low sampling efficiency, DAPO uses a generation batch size that is 1–4 times the training batch size and adjusts it dynamically. Across the four models, DAPO requires 3 to 5 times more generated responses than DCPO for the same number of parameter updates, at least doubling training GPU hours without better performance. Its utilization also decreases with model size; for Qwen2.5-14B, fewer than 30% of responses are used after about 300 steps. This shows that dynamic sampling can reduce data utilization and leave less opportunity to explore response diversity.

Overall, DCPO improves over GRPO and DAPO in this response-utilization analysis by retaining more generated samples for optimization, reducing wasted training compute, and preserving more opportunity for diverse reasoning exploration while improving benchmark performance.

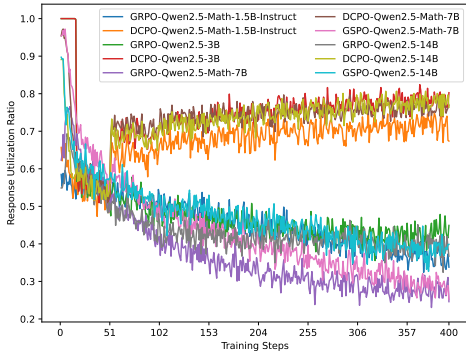


Figure 2: Response utilization ratio through-out training.

5.4 ABLATION STUDIES

We conduct an ablation study on Qwen2.5-Math-7B to isolate the contribution of each DCPO component under the same mathematical RLVR setup, using **Avg@32** as the metric because it reflects the robustness and stability of the learned policy distribution under sampling. For fairness, each ablation changes only one component of the GRPO baseline while holding all other settings fixed and removing the KL divergence term to match DAPO, GSPO, and full DCPO. Figure 3 reports the learning curves from 20 to 400 steps.

The curves show that each individual component improves inference ability over training and can surpass GRPO while reaching or exceeding DAPO and GSPO in this setting. The components therefore address complementary parts of the RLVR optimization problem, and their combination yields the largest gain over the baselines. In later training, baseline performance plateaus, whereas DCPO continues to improve both as a full method and at the component level. This supports the interpretation that the gains come from improved data use, probability-aware exploration, and more stable response-level optimization rather than from a single isolated change.

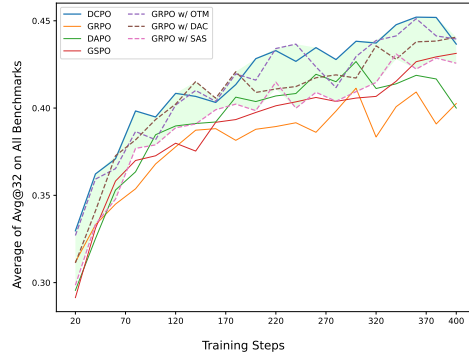


Figure 3: Avg@32 ablation curves on Qwen2.5-Math-7B.

(i) **Only-Token-Mean loss (OTM)**: Aggregating loss within each response rather than across the entire batch improves over baselines that suffer from weaker data utilization and larger performance fluctuations. (ii) **Smoothed Advantage Standardization (SAS)**: Replacing stepwise advantages with cumulative smoothed advantages substantially improves GRPO and approaches DAPO, showing that prompt-level reward statistics across steps can stabilize optimization. (iii) **Dynamic Adaptive Clipping (DAC)**: Replacing GRPO’s fixed clipping boundary with a probability-adaptive boundary yields large gains; DAC alone surpasses GRPO, DAPO, and GSPO and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: Combining the three mechanisms gives the strongest and most stable results, indicating that response-level aggregation, cumulative reward smoothing, and token-probability-aware clipping each contribute to data-efficient and stable RLVR training.

Overall, the ablation confirms that the DCPO components provide complementary improvements and that their combination produces cumulative gains in the evaluated model and benchmark setting. The results support the broader mechanism-level claim that better response utilization and probability-aware token updates improve data efficiency and stability for RLVR with LLMs.

6 CONCLUSION

This work addresses two limitations that arise broadly in RLVR policy optimization: fixed clipping restricts token-level exploration in low-probability regions, and current-step reward standardization can lower sample efficiency by producing zero advantages. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, an RLVR optimization pipeline with two central mechanisms: (i) **dynamically adaptive token-level clipping**, which sets bounds from old-policy token probabilities so rare tokens can be explored more effectively while common tokens remain tightly controlled, and (ii) **Cumulative-Smooth Advantage Standardization**, which incorporates cumulative prompt-level reward statistics across optimization steps to reduce instability from high-entropy sampling. Experiments on four mathematical reasoning benchmarks and four model settings demonstrate the effectiveness of this framing within the evaluated scope. For example, DCPO-7B reaches 38.8 Avg@32 on AIME24, outperforming GRPO (32.1) and DAPO (31.6). DCPO also improves the response utilization ratio by 28% over GRPO, doubles training efficiency compared with DAPO, and reduces the token clipping ratio by an order of magnitude. These results indicate that more general probability-aware clipping and cumulative advantage smoothing can improve data utilization and diversity exploration in verifiable-reward training. Future work will test whether the same mechanisms transfer to other verifiable domains, such as code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In GRPO, DAPO, and related methods, the token advantage $\hat{A}_{j,t}^i$ for token t in response j is obtained by standardizing the response reward R_j^i with the mean μ^i and standard deviation σ^i of the G responses generated for the same prompt at step i , as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When all rewards for a prompt are identical, the resulting zero advantages prevent those responses from contributing to the update, which wastes generated data.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses for each query, assigns rule-based rewards R , and estimates token-level advantages as in Equation (11). The parameters are then updated with Equation (12), the **Sequence Level Mean loss (SLM)**:

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level advantages within each response and then across all G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty keeps the policy close to a reference policy (e.g., a pre-trained model), while the fixed clipping threshold ε limits large changes in token probability ratios. Although this structure is broadly used, its batch averaging can dilute response-level importance.

A.3.2 DAPO

Yu et al. (2025) propose **Dynamic sAmpling Policy Optimization (DAPO)** to address the observation that longer GRPO sequences can have disproportionate gradient influence. DAPO normalizes token counts over the full batch, yielding the **Token Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. Prompts whose responses all share the same reward are discarded and regenerated to maintain batch size, improving gradient availability at the cost of substantial data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) propose **Group Sequence Policy Optimization (GSPO)**, arguing that token-level clipping can add high-variance noise to gradient estimates. Because this noise can accumulate over long sequences and be amplified by clipping, GSPO replaces token-wise clipping with sequence-wise clipping to produce a more stable training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

where $s_i(\theta) = \exp\left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log\left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)}\right)\right)$

In the evaluated settings, GSPO discards more than 10% of responses with nonzero advantages, which reduces data utilization and training efficiency. Although sequence-level clipping removes some high-variance responses, many token-level high-entropy signals remain, so training instability can still arise.

A.4 REWARD CALCULATION

To keep supervision verifiable and interpretable, we use a rule-based reward model that scores each generated response by answer correctness and format compliance, as in Equation (15). A response receives +1 when both format and answer are correct, 0 when the format is correct but the answer is wrong, and -1 when the format is incorrect.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response group for a given problem at training step i , and the reward depends only on response format and answer correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Probability-ratio clipping is a general reinforcement-learning tool for stabilizing policy optimization and preventing large parameter updates. It is especially important when importance sampling estimates expectations under a new policy from samples generated by an old policy.

The expectation of a function $f(x)$ under the new probability $p(x)$ can be expressed under the old probability $q(x)$ through an importance-sampling weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, it can have much higher variance, a general challenge for importance sampling. The variance of $f(x)$ under the new policy is given in Equation (17), while the variance of the importance-sampled estimator under the old policy is given in Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} [f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x) \frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The gap between the importance-sampled variance and the new-policy variance makes this inflation explicit, as shown in Equation (19):

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To reduce this variance gap, many methods, including PPO Schulman et al. (2017) and RLVR approaches (DeepSeek-AI et al., 2024; Liu et al., 2025), use a fixed symmetric clipping bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed clipping bound, however, constrains exploration most strongly where the old policy assigns low probability mass. In those regions, the admissible absolute change is limited, reducing the chance of meaningful updates on rare tokens. Asymmetric methods such as DAPO relax this constraint on one side, but the bound is still not adapted to token probability.

Recent studies such as Yue et al. (2025) also suggest that many reasoning paths used by RLVR models already lie within the base model’s sampling distribution. The small fraction of clipped parameters observed during off-policy batch updates further indicates limited parameter shift. This motivates a probability-aware alternative: we constrain the ratio $r(x)$ with a dynamic-adaptive restriction that includes the probability term, as in Equation (21), rather than relying only on the fixed bound in Equation (20).

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality yields the expression in Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For the clipping ratio to remain valid, it must satisfy the standard constraints on probabilities and probability ratios. As shown in Equation (23), $p(x)$ and $q(x)$ are non-negative and at most 1, so $r(x)$ and the clipping hyperparameter ϵ must also be non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

To make the bound derived from Equation (22) usable in policy optimization, we define the necessary bound conditions in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2} \sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) and Equation (24) gives the following admissible conditions for the probability ratio $r(x)$:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

For clipping, we use the overlapping region centered at ratio 1. This minimizes discrepancies between the two cases and keeps the resulting bound aligned with the conventional GRPO fixed bound:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, we keep the square-root term non-negative by comparing with 0 and taking the maximum, such as $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This yields the following **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2} \sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we use separate hyperparameters ϵ_{high} and ϵ_{low} so the dynamic bound aligns with GRPO at selected reference points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To compare dynamic-adaptive clipping with fixed symmetric clipping on the same reference scale, we align the dynamic bounds with the fixed-bound points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d. Solving Equation (28) gives the corresponding values of ϵ_{low} and ϵ_{high} .

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \\ q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases} \quad (28)$$

Using the common fixed threshold $\epsilon = 0.2$ from prior practice such as GRPO gives dynamic hyperparameters of approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although $\epsilon_{low} \neq \epsilon_{high}$, this differs from DAPO-style asymmetric fixed clipping because the resulting bounds still vary with token probability.

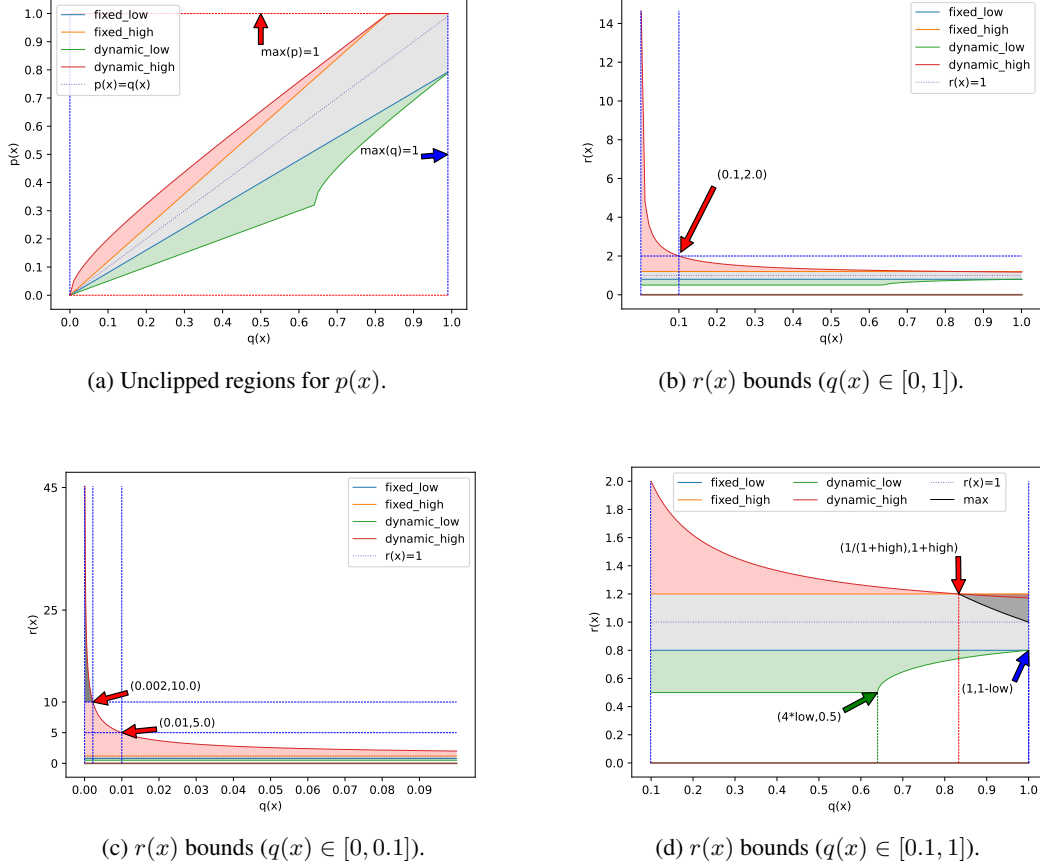


Figure 4: Comparison of fixed clipping ($\epsilon = 0.2$) and dynamic-adaptive clipping ($\epsilon_{low} = 0.16$, $\epsilon_{high} = 0.2$) across probability ranges.

Dynamic versus fixed clipping: Figure 4 contrasts the fixed GRPO clipping interval $[1 - \epsilon, 1 + \epsilon]$ with probability-aware adaptive clipping. For a token with old probability $q(x)$, the fixed bound always has relative width 2ϵ , so the absolute exploration range shrinks linearly as $q(x)$ decreases (Figure 4a). Dynamic adaptation instead makes the relative width inversely related to $q(x)$, keeping the effective exploration space more consistent across token probabilities.

Effect of decreasing $q(x)$: When $q(x)$ is small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains unchanged and therefore does not increase the model’s capacity to explore low-probability tokens. The dynamic-adaptive bound expands rapidly in this region (Figure 4b), giving rare tokens a larger relative update space while keeping the update rule controlled.

Behavior for extremely low probabilities: Figure 4c focuses on the low-probability range $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$ grows rapidly with the inverse square-root form, whereas the fixed-clipping bound remains constant. Following the dual-clipping strategy (Ye et al., 2020), we impose a hard ceiling $r_{max} = 10$ to avoid instabilities such as gradient vanishing or explosion. The gray region in Figure 4c corresponds to approximately $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is capped at r_{max} .

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\epsilon$, dynamic-adaptive clipping gives $dynamic_{low} = 0.5$, while fixed clipping keeps $fixed_{low} = 1 - \epsilon = 0.8$. Above this threshold, the dynamic lower bound smoothly contracts toward the fixed value. Likewise, the dynamic upper

bound exceeds the fixed bound when $q(x) < \frac{1}{1+\epsilon_{\text{high}}}$ and merges with it for larger $q(x)$. The interval $[dynamic_{\text{low}}(q(x), \epsilon_{\text{low}}), dynamic_{\text{high}}(q(x), \epsilon_{\text{high}})]$ therefore widens the admissible region for rare tokens while preserving a tight interval for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) includes both positive- and negative-advantage cases. For positive advantages, the minimizing operation clips $r(x)$ within $[0, dynamic_{\text{high}}(q(x), \epsilon_{\text{high}})]$; for negative advantages, it uses $[dynamic_{\text{low}}(q(x), \epsilon_{\text{low}}), \infty)$. Because very large ratios can destabilize training, we cap the ratio for both cases at $r_{\text{max}} = 10$, following the motivation of Ye et al. (2020), so the adaptive bounds do not create excessive importance weights.

In summary, for both positive and negative advantages, dynamic-adaptive clipping gives rare low-probability tokens more exploration space than fixed clipping while preserving bounded updates.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the training iteration in which prompts generate responses and update model parameters. The following statistics define reward normalization over current and cumulative responses.

The response-reward means are defined in Equation (29):

$$\begin{cases} \mu_{\text{new}}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{\text{old}}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{\text{total}}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{\text{new}}^i + (i-1) \mu_{\text{old}}^i) \end{cases} \quad (29)$$

The corresponding response-reward variances are defined in Equation (30):

$$\begin{cases} \sigma_{\text{new}}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{\text{new}}^i)^2} \\ \sigma_{\text{old}}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{\text{old}}^i)^2} \\ \sigma_{\text{total}}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{\text{total}}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{\text{new}}^i{}^2 + (i-1) \sigma_{\text{old}}^i{}^2 + \frac{i-1}{i} (\mu_{\text{old}}^i - \mu_{\text{new}}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce resource usage when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent calculations based on μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than recomputing from all original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments use the following prompt format, based on the Qwen-Math template (Yang et al., 2024b), so comparisons share the same instruction and answer-format assumptions.

Qwen-Math Template

```

<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant

```

A.9 TRAINING SETTING

For GRPO, we use a symmetric clipping threshold $\epsilon = 0.2$. For DAPO, we use lower/upper clipping thresholds $(0.2, 0.28)$, an additional 512-token soft-punish cache, and maximum probability ratio 10 for negative advantages. For GSPO, we use lower/upper clipping thresholds $(3e-4, 4e-4)$. For DCPO, we use low/high thresholds $(0.16, 0.2)$, preserving the same reference clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1+\epsilon)$ and $(q(x), r(x)) = (1, 1-\epsilon)$ as GRPO (Figure 4d), and set the maximum probability ratio to 10 for both positive and negative advantages, unlike DAPO.

Training runs for 400 steps with response-generation batch size 512 and parameter-update mini-batch size 32. We generate $G=16$ responses per problem with temperature 1 and top_p 1. The maximum prompt length is 1,024 and the maximum response length is 3,072, which is much shorter than the 20k maximum length used in the original DAPO setting.

We adapt the training codebase from Verl (Sheng et al., 2024) and run experiments on 32xH20 GPUs. Unless otherwise specified, hyperparameters match DAPO. For loss computation, we keep the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulation from GRPO or DAPO rather than the approximate weighted calculation used in Verl, ensuring compatibility across microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

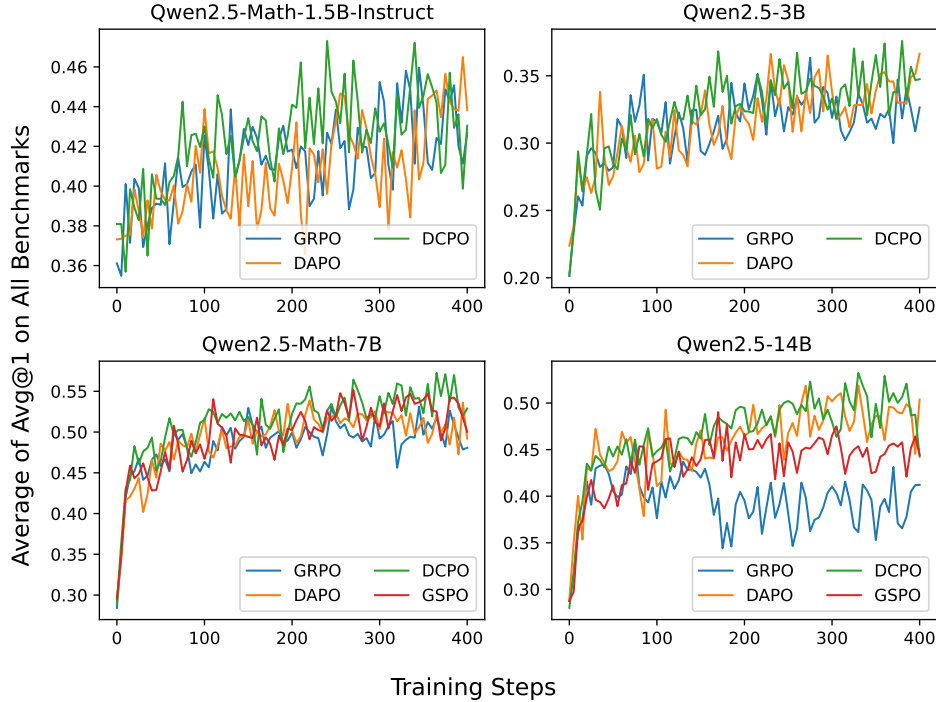


Figure 5: Avg@1 performance across the evaluated benchmarks.

Avg@1 Trend Figure 5 reports the average Avg@1 metric, computed by averaging greedy-decoding results over MATH500, AMC23, AIME24, and AIME25. Across the four evaluated models, DCPO consistently outperforms GRPO and DAPO, and it outperforms GSPO on the two models where GSPO is included. Its trajectory is also more stable than the baselines, avoiding the pronounced performance drop observed for GRPO and the larger fluctuations observed for DAPO. This indicates that DCPO improves deterministic reasoning performance while maintaining a steadier optimization path across model scales.

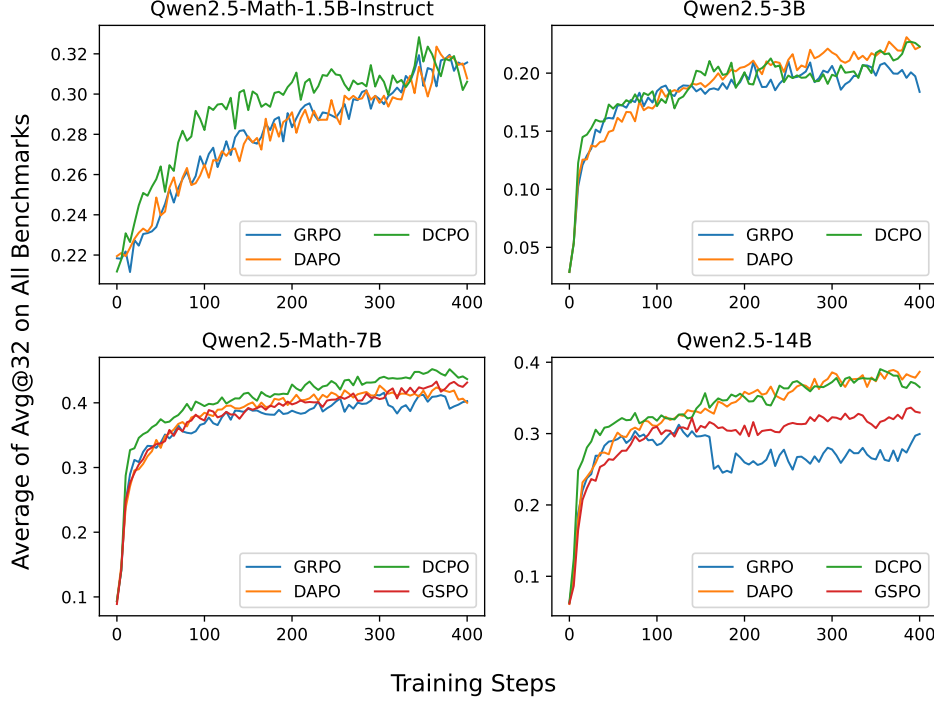


Figure 6: Avg@32 performance across the sampled-evaluation benchmarks.

Avg@32 Trend Figure 6 compares Avg@32 performance for DCPO, GRPO, DAPO, and, where available, GSPO across model sizes on AMC23, AIME24, and AIME25. GRPO shows substantial fluctuation and occasional collapse; on Qwen2.5-14B, it drops sharply after roughly 170 training steps. DCPO instead follows a generally upward trajectory with smaller fluctuations across the evaluated 1.5B–14B range, including faster early-stage growth. Compared with DAPO, DCPO attains stronger performance while using roughly half the training time and resources. These results support the claim that dynamic-adaptive clipping and smoothed advantage standardization improve response-level exploration and label-diversity utilization under sampled decoding.

Entropy Trend Figure 7 shows that GRPO undergoes sharp entropy decay after an initial warm-up and then remains in a flat, low-entropy regime. This collapse indicates a rapid loss of policy randomness, reducing sampled-response diversity and limiting later policy optimization.

DAPO and DCPO show model-dependent entropy behavior. On the Instruct model (Qwen2.5-Math-1.5B), DAPO, DCPO, and GRPO follow similar declining trajectories. On the larger base models (Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B), the trends separate: DAPO maintains the highest entropy with a clear upward trend and large fluctuations. On Qwen2.5-Math-7B, GSPO has entropy similar to DCPO and comparable average metric performance, whereas on Qwen2.5-14B, GSPO performance fluctuates downward. DCPO remains in a moderate entropy band, smoother than DAPO and higher than the low-entropy GRPO regime.

These observations suggest a U-shaped relationship between entropy and training performance in the evaluated runs. Too little entropy removes exploratory signal and encourages premature convergence, while too much or highly unstable entropy harms training stability. DCPO keeps entropy in a

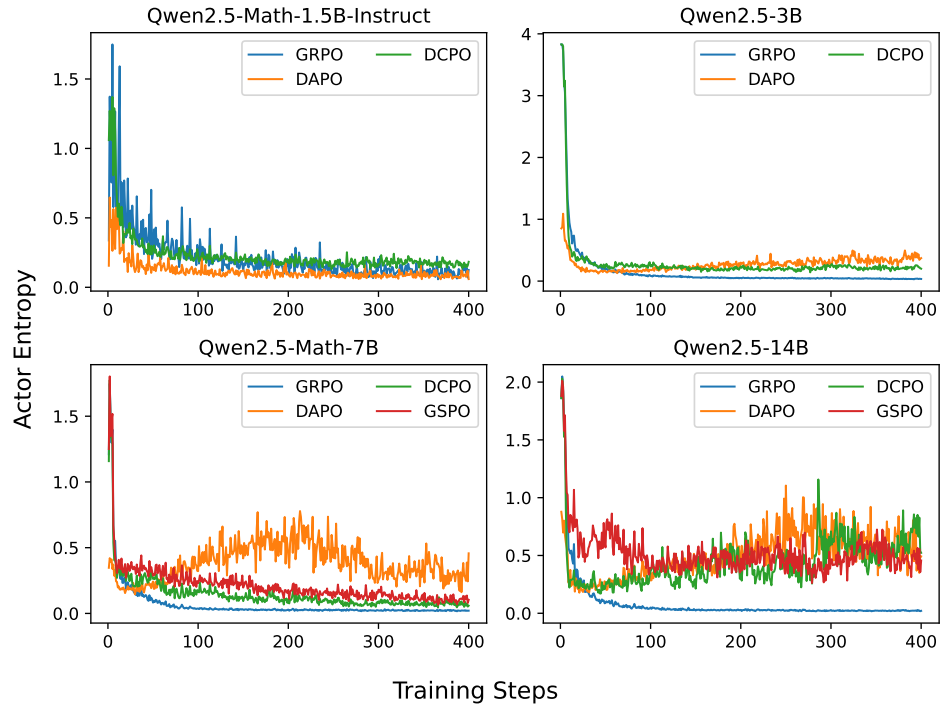


Figure 7: Model entropy trajectories during training.

moderate range, supporting both convergence and exploration; this balance is especially visible as model capacity increases and may help explain DCPO’s stronger inference performance across the tested model sizes.